Meta | Startupbootcamp | أكاديمية طويق Tuwaiq Academy

Project 3

An AI-Powered Diabetes Care and Follow-Up Assistant

Adeed | عضيد



Team Members:
Shahad Al-Muhaisin.

Bootcamp Instructors: Eng. Lulwah Al-Yahya & Eng. Turki Al-Balawi.

Executive Summary

Diabetes management involves more than collecting glucose readings. Patients may need to understand technical report metrics, compare results across time, organize relevant laboratory information, and prepare questions for clinical appointments. A digital service that brings these tasks into one structured workflow may help patients engage with their information more confidently. However, the extent of this need among the intended users must be validated rather than assumed.

Adheed is a proposed AI-assisted diabetes-care and follow-up service designed to fit within a Sehhaty-inspired digital health experience. The concept combines supported glucose-report extraction, data validation, comparison with compatible historical results, conversational explanations, appointment preparation, and a structured pathway for professional review.

The International Diabetes Federation estimates that 5.3 million adults aged 20-79 were living with diabetes in Saudi Arabia in 2024, representing an age-standardised prevalence of 23.1% (International Diabetes Federation [IDF], 2025). This establishes the public-health relevance of exploring practical digital support, but it does not by itself prove the specific usability problem addressed by Adheed.

The proposed design deliberately separates deterministic processing from generative AI. Numerical calculations, unit checks, historical comparability, and workflow state changes should be handled by tested application logic. AI should be limited to explaining verified information, answering general educational questions, and helping patients prepare questions for a clinician. It must not diagnose, prescribe insulin doses, or independently change treatment.

Adheed is a concept and prototype proposal, not an officially endorsed Ministry of Health or Sehhaty feature. Its value should be assessed through problem validation, extraction and comparison testing, usability studies, safety evaluation, and feedback from patients and healthcare professionals.

1. Target Organization

1.1 Organization Overview

The Saudi Ministry of Health (MOH) provides healthcare services and supports digital health services in the Kingdom. The Ministry describes Sehhaty as a unified platform through which users can access health information and services, including appointment-related services, medical reports, remote consultations, and selected health and wellness

1.2 Rationale for Selection

Diabetes care can involve repeated glucose monitoring, laboratory testing, and discussions with healthcare professionals. A service that organizes selected results and helps patients prepare for follow-up is therefore a relevant design opportunity. The IDF's 2025 Diabetes Atlas reports an estimated 5.3 million adults with diabetes in Saudi Arabia in 2024, highlighting the scale of the condition (IDF, 2025). The proposed project focuses on one defined workflow rather than redesigning the entire platform.

1.3 Existing Service Opportunity

The opportunity explored by this project is a connected workflow for reviewing glucose reports, comparing compatible historical measurements, preparing questions, and navigating professional follow-up. Since Sehhaty already provides several digital health services, the project must not claim that appointment booking, medical reports, or consultations are absent. Instead, the specific gap-whether patients need a more consolidated glucose-report interpretation and preparation workflow-should be tested through official service review and user research.

2. Industry Background

Diabetes self-management may involve information from continuous glucose monitoring (CGM), insulin-pump or sensor reports, laboratory tests such as HbA1c, and previous care recommendations. These data sources are not interchangeable: they may cover different time periods, use different units, and measure different aspects of glycaemia.

The American Diabetes Association's Standards of Care discuss CGM metrics such as time in range, time above range, and time below range as useful elements of glycaemic assessment. They also describe the importance of sufficient CGM data when interpreting these measures (American Diabetes Association Professional Practice Committee, 2025). Adeed should therefore display each metric with its date range, unit, source, and any relevant data-quality limitation.

AI may support document understanding and plain-language explanations, but healthcare applications require safeguards, accountability, transparency, and human oversight. The World Health Organization's guidance on ethics and governance of AI for health emphasizes that ethical principles and human rights should be central to the design and use of AI systems (World Health Organization [WHO], 2021). Accordingly, Adeed is designed as an informational and coordination aid-not an autonomous clinical decision-maker.

3. Problem Validation

3.1 Problem Context

People managing diabetes may need to interpret glucose sensor or insulin-pump reports, understand changes in their readings over time, and prepare questions for clinical consultations. These tasks can involve several metrics, different reporting periods, and laboratory measurements that should not always be interpreted in the same way.

The proposed Aḍeed feature addresses this potential challenge by organizing supported glucose-report metrics, comparing compatible historical results, and helping patients prepare for conversations with healthcare professionals.

However, the relevance of diabetes as a public-health issue does not, by itself, establish that patients experience this specific difficulty. The proposed user problem must therefore be examined using credible research and, where feasible, direct feedback from intended users.

3.2 Evidence and Its Limits

The following sources provide context for the project and help establish why the proposed problem is worth investigating.

Evidence 1: Diabetes burden in Saudi Arabia

The International Diabetes Federation's Diabetes Atlas provides country-level estimates of diabetes prevalence and population burden. This information establishes the public-health relevance of exploring services that support people living with diabetes.

Relevance to Aḍeed: The scale of diabetes in Saudi Arabia provides a rationale for investigating whether a focused digital service could help patients understand and organize their glucose-report information.

Limitation: Prevalence statistics do not demonstrate that patients struggle to interpret reports, compare historical results, or prepare questions for appointments.

Evidence 2: Complexity of glucose-monitoring metrics

The American Diabetes Association's Standards of Care discuss the use of continuous glucose monitoring metrics, including time in range, time above range, and time below range, in diabetes assessment.

Relevance to Aḍeed: These metrics provide a clinical context for the proposed feature's plain-language explanations and structured report presentation.

Limitation: Clinical guidance establishes the relevance of these measurements, but it does not prove that patients find them difficult to understand.

Evidence 3: Digital healthcare and AI governance

The Saudi Ministry of Health provides digital healthcare services through Sehhaty. Separately, the World Health Organization's guidance on AI for health emphasizes the importance of responsible governance, transparency, and human oversight.

Relevance to Aḍeed: These sources inform the proposed digital-service context and support the need to design AI assistance with appropriate safeguards.

Limitation: Neither source establishes that the exact Aḍeed workflow is missing from current services or that users would adopt it.

3.3 Evidence Gap and Further Validation

The available secondary evidence establishes the relevance of diabetes care and the importance of presenting health information responsibly. However, it does not yet establish the frequency or severity of the specific user problem addressed by Aḍeed.

Further validation should investigate how intended users currently access glucose reports, interpret their metrics, compare previous results, and prepare for appointments.

If feasible, the team should conduct a small exploratory study involving people who manage diabetes and healthcare professionals. Participants could be asked to complete a report-interpretation task using a de-identified sample report. The team could then document observed difficulties, questions raised, task completion, and participant feedback.

Any interviews or usability tests must be reported only after they have actually been conducted. The report should identify the research method, participant count, recruitment approach, findings, and limitations. A small convenience sample should not be presented as representative of the wider population.

Until this additional evidence is collected, the specific user problem should be treated as a research hypothesis rather than an established finding.

3.4 Problem Statement

People managing diabetes who use glucose sensor or insulin-pump reports may need a more convenient way to understand technical metrics, compare compatible historical results, and prepare focused questions for clinical consultations. Aḍeed proposes a guided digital workflow that brings these activities together and explains verified information in accessible language.

The project will evaluate whether this approach improves users' understanding and appointment preparation. The extent of the underlying need and the effectiveness of the proposed solution remain to be validated through user research and usability testing.

4. Target Users

4.1 Primary Users: People Living with Diabetes

The primary users are people who receive supported glucose sensor or insulin-pump reports and want help organizing information before a clinical appointment.

• Understand the meaning of supported report metrics in accessible language.

• Review current results alongside compatible previous reports and laboratory

measurements. • See when historical data is missing, incomplete, or unsuitable for

comparison.

• Prepare questions for a healthcare professional.

4.2 Secondary Users: Healthcare Professionals

The secondary users are authorized healthcare professionals who may review patient-submitted reports and need a structured view of relevant results, historical changes, and data limitations.

• Review source information and data-quality warnings.

• Understand the reason for a patient's review request.

• Document recommendations or request additional information.

• Recommend appropriate follow-up or laboratory testing within their professional scope.

4.3 Other Stakeholders

Potential stakeholders include healthcare organizations, laboratory providers, digital health product teams, privacy and security teams, and administrators responsible for clinical governance. Their involvement would depend on the implementation and any future integrations.

5. Proposed AI Solution

5.1 Solution Overview

Adheed is a proposed AI-assisted workflow for organizing glucose reports, comparing compatible historical measurements, explaining verified information, and supporting preparation for professional follow-up. The patient-facing prototype places the Adheed entry point inside the My Profile area of a Sehhaty-inspired interface. This is a design choice for the prototype, not a claim about an official Sehhaty implementation.

5.2 Proposed Workflow

The patient opens Adheed and uploads a supported glucose sensor or insulin-pump PDF report. The

application checks the file type and size, extracts supported fields, and validates values and units.

Each extracted value is linked to its source label and page where technically possible; uncertain or unavailable fields are explicitly marked.
The application retrieves only historical records belonging to the authenticated patient.

Deterministic logic compares compatible measurements and displays dates, reporting periods, units, differences, and comparability warnings.

The patient reviews the structured summary and can ask the assistant to explain validated metrics or help prepare questions.

When appropriate, the patient submits a review request through the defined workflow.

An authorized clinician reviews the current report, relevant history, source information, and limitations, then records a recommendation or requests additional information.

If follow-up or laboratory testing is recommended, the patient can use a verified booking integration where one exists. A simulated prototype must be labelled clearly and must not imply a real reservation.

5.3 Why AI Is Appropriate-and Where It Is Not

AI is potentially useful for interpreting a patient's natural-language question, explaining technical terms in plain language, and assisting with variable document layouts when conventional extraction is insufficient. It can also help turn verified report information into a readable appointment-preparation summary.

AI should not be responsible for arithmetic, unit conversion without validation, deciding whether two measurements are comparable, enforcing permissions, or changing review and booking states. These tasks should use deterministic code and testable rules. The assistant must use only verified information available to the authenticated user, identify uncertainty, and avoid inventing values, diagnoses, clinical recommendations, or appointments.

The system must not diagnose a condition, recommend insulin doses, or independently change medication or treatment. Clinical decisions remain with qualified healthcare professionals. The product should also communicate that it is not an emergency service and should direct users to appropriate urgent care when a message indicates a potentially urgent situation, without delaying access to care.

6. Core Features

6.1 Glucose Report Extraction

Accept supported PDF reports, validate uploads, extract available fields, and preserve source labels, units, and page references where possible. Show fields as extracted, calculated, unavailable, or uncertain. This reduces manual effort while making limitations visible.

6.2 Historical Results Comparison

Compare the current report with the latest compatible previous glucose reports and relevant laboratory results. Depending on report availability, metrics may include average glucose, time in range, time above range, time below range, glucose variability, and GMI. Display dates, periods, units, previous and current values, and differences. HbA1c is a laboratory measurement and GMI is an estimate; they must be displayed separately and never treated as interchangeable.

6.3 AI Patient Assistant

Answer general diabetes-education questions, explain validated metrics, summarize available historical comparisons, and help users prepare questions for a clinician. Responses should be grounded in verified data and should acknowledge missing information. The assistant must not diagnose or prescribe treatment.

6.4 Appointment Preparation Summary

Consolidate the current report, compatible historical comparisons, relevant available laboratory results, missing information, and suggested questions. The summary should help structure a consultation without claiming to replace clinical judgment.

7. User Journey

7.1 Current Journey (Research Hypothesis)

The following is a preliminary journey hypothesis and should be replaced or refined using evidence from user research or verified service observation.

The patient receives or downloads a glucose report.

The patient attempts to understand the report's technical metrics.

The patient locates previous glucose reports or laboratory results, if available.

The patient tries to identify meaningful changes across the records.

The patient prepares questions or concerns before a clinical appointment.

The patient discusses the information with a healthcare professional.

Potential friction points to investigate include unfamiliar terminology, manual comparison, missing historical data, uncertainty about whether measurements are comparable, and difficulty organizing questions. These should not be reported as confirmed findings unless research supports them.

7.2 Proposed Adeed Journey

The patient opens Adeed from the My Profile section of the prototype.

The patient uploads a supported report and sees the processing status.

The application validates the report and presents available metrics with source information.

Compatible historical results are shown beside the current results, with dates, units, and

warnings. The AI assistant explains selected information and helps the patient draft questions.

The patient submits a review request when appropriate.

An authorized clinician reviews the submitted information and records a decision or requests additional details.

The patient sees the recorded recommendation and, where implemented, accesses the relevant follow-up or laboratory booking workflow.

The patient can return to previous reports and review history.

7.3 Journey Comparison

8. Wireframes and Product Screens



9. Expected Business and User Impact

9.1 Expected User Benefits

If the underlying need is validated and the system performs reliably, potential benefits include easier access to report information, more understandable historical comparisons, reduced effort in organizing results, better-structured appointment preparation, and clearer visibility into review status. These are expected benefits, not proven outcomes.

9.2 Expected Organizational Benefits

Potential organizational benefits include more structured information submitted for review, better visibility into pending requests, and a more organized follow-up workflow. Real impact would depend on adoption, clinical review capacity, data quality, integration quality, and governance.

9.3 Success Metrics

Numerical targets should be agreed before evaluation and justified by the project scope. Do not report a target as an achieved result. Any claims about clinical effectiveness require appropriate clinical validation and are outside what a basic course prototype can establish.

10. Limitations and Risk Management

The prototype should not be presented as a clinically validated product. Any real-world deployment would require further security, privacy, clinical governance, regulatory, and integration review.

11. Conclusion

Adeed proposes a focused AI-assisted workflow for organizing glucose reports, comparing current measurements with compatible historical results, explaining verified information, and supporting professional follow-up. The concept responds to a relevant public-health context while recognizing that the specific user problem still needs direct validation.

Its central design principle is to combine deterministic data processing with constrained AI assistance and human clinical oversight. By making sources, units, missing data, and comparison limitations visible, the proposed service aims to support clearer patient understanding without replacing medical judgment. The next steps are to validate the problem with intended users, test the report-processing workflow, evaluate usability and safety, and verify any integration assumptions.

12. References

American Diabetes Association Professional Practice Committee. (2025). 7. Diabetes technology: Standards of Care in Diabetes-2025. Diabetes Care, 48(Supplement 1). https://diabetesjournals.org/care/article/48/Supplement_1/S145/157557

International Diabetes Federation. (2025). IDF Diabetes Atlas (11th ed.): Saudi Arabia country report. https://diabetesatlas.org/data-by-location/country/saudi-arabia/

Ministry of Health, Kingdom of Saudi Arabia. (n.d.). Sehhaty platform. https://www.moh.gov.sa/en/support/pages/sehhatyapp.aspx

World Health Organization. (2021). Ethics and governance of artificial intelligence for health: WHO guidance. https://www.who.int/publications/i/item/9789240029200